

Classical Mechanics 1, Autumn 2025 CMI

Assignment 7

Due by 6pm, 25 Oct, 2025

Simple plane pendulum

1. **⟨9⟩ Simple pendulum: angular momentum and torque.** Consider a simple pendulum with bob of mass  $m$  and rod of length  $\ell$  subject to Earth's acceleration due to gravity with magnitude  $g$ . Denote the counterclockwise deflection angle  $\theta$  and use the coordinate frame developed in the lecture.
  - (a) **⟨3⟩** Find an expression (in terms of  $\theta$ ) for the torque  $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$  on the bob about the pivot. Which forces do/do not contribute to  $\boldsymbol{\tau}$ ?
  - (b) **⟨1⟩** Write an expression (in terms of  $\theta$ ) for the angular momentum  $\mathbf{L} = \mathbf{r} \times \mathbf{p}$  of the bob about the pivot.
  - (c) **⟨2⟩** Use the evolution equation for angular momentum to derive a differential equation for  $\theta(t)$ . How does it compare with the EOM for  $\theta$  derived from Newton's second law?
  - (d) **⟨3⟩** Propose a limit holding  $m, \ell, g$  fixed in which the angular momentum is asymptotically conserved. Specify the limit through a suitable comparison of physical quantities. Comment on the nature of motion in this limit.
2. **⟨11⟩ Linearization of pendulum equation.** Linearize the equation of motion  $\ddot{\theta} = -\omega^2 \sin \theta$  for a simple pendulum around  $\theta = \pi$ . Here  $\omega = \sqrt{g/\ell}$ .
  - (a) **⟨2⟩** Find the approximate linear equation satisfied by  $\phi = \theta - \pi$ .
  - (b) **⟨4⟩** Use a suitable guess to find two linearly independent solutions of the equation for  $\phi$  and the general solution of the equation for given  $\phi(0)$  and  $\dot{\phi}(0)$ .
  - (c) **⟨2⟩** Physically interpret the consequences of the solution of the linear differential equation for  $\phi$  for typical (nonexceptional) initial conditions and comment on what it means for the stability of the static solution  $\theta = \pi$ .
  - (d) **⟨3⟩** Comment on any exceptional ICs and the resulting time evolution.